DFS Connectivity Check: Solution

Emirates Post Network Audit · Design and Analysis of Algorithms · Data Lead: Kushal

1. Problem

Emirates Post runs parcels from its central hub at Umm Ramool (HUB) to six post offices. Google Maps gives a drive time between every pair (21 pairs), but a real network only runs short corridors. We keep a direct route only if its drive time is **under X minutes**, then use **Depth-First Search (DFS)** to check that every branch can still be reached from the hub.

Code	Post office	Area
HUB	Emirates Post - Al Ramool Post Office (hub)	Umm Ramool, Dubai
KAR	NXN - Dubai Central	Al Karama, Dubai
QUS	Emirates Post - Al Qusais Post Office, 5th St	Al Qusais, Dubai
UMS	NXN - Umm Suqeim Post Office	Umm Suqeim, Dubai
JAF	Emirates Post - JAFZA Post Office, Expo Rd	Jebel Ali Free Zone, Dubai
SHJ	NXN - Sharjah Central Post Office, Government House Square	Al Soor, Sharjah
AJM	Emirates Post - Ajman Central Post Office, Masfout St	Al Bustan, Ajman

Data: Google Maps driving times, depart Thursday 5 Nov 2026, 07:00 (morning van run). Minutes = midpoint of Google's typical range for the fastest route, rounded. Each pair is used in both directions.

2. Symmetric time matrix (minutes)

	HUB	KAR	QUS	UMS	JAF	SHJ	AJM
HUB	0	**13**	**16**	**23**	38	**24**	38
KAR	**13**	0	**25**	**20**	**34**	**30**	48
QUS	**16**	**25**	0	**32**	48	**29**	**37**
UMS	**23**	**20**	**32**	0	**24**	40	55
JAF	38	**34**	48	**24**	0	53	65
SHJ	**24**	**30**	**29**	40	53	0	**27**
AJM	38	48	**37**	55	65	**27**	0

Green = under 38 minutes, so a route exists. The matrix is symmetric (row A, column B = row B, column A).

3. Choosing the threshold X = 38 minutes

X (min)	Routes under X	10-14 routes?	Connected?	Cut off
27	7	no	no	AJM
28	8	no	YES	-
30	9	no	YES	-
33	11	yes	YES	-
35	12	yes	YES	-
38	**13**	**yes**	**YES**	**-**
39	15	no	YES	-

Why 38: it is the hub's direct drive to its two farthest offices (Jebel Ali, Ajman), so no corridor is longer than the hub's own longest run. It is the largest X inside the rule (X = 39 gives 15 routes). 13 routes, all connected, and still connected at X = 33. A route needs a time strictly under 38, so HUB-JAF and HUB-AJM (38) are excluded.

4. Step 1: build the neighbour lists (route exists if time < 38)

Branch	Check each time against 38	Neighbours (matrix order)	Count
HUB	KAR 13✓ · QUS 16✓ · UMS 23✓ · JAF 38✗ · SHJ 24✓ · AJM 38✗	KAR, QUS, UMS, SHJ	4
KAR	HUB 13✓ · QUS 25✓ · UMS 20✓ · JAF 34✓ · SHJ 30✓ · AJM 48✗	HUB, QUS, UMS, JAF, SHJ	5
QUS	HUB 16✓ · KAR 25✓ · UMS 32✓ · JAF 48✗ · SHJ 29✓ · AJM 37✓	HUB, KAR, UMS, SHJ, AJM	5
UMS	HUB 23✓ · KAR 20✓ · QUS 32✓ · JAF 24✓ · SHJ 40✗ · AJM 55✗	HUB, KAR, QUS, JAF	4
JAF	HUB 38✗ · KAR 34✓ · QUS 48✗ · UMS 24✓ · SHJ 53✗ · AJM 65✗	KAR, UMS	2
SHJ	HUB 24✓ · KAR 30✓ · QUS 29✓ · UMS 40✗ · JAF 53✗ · AJM 27✓	HUB, KAR, QUS, AJM	4
AJM	HUB 38✗ · KAR 48✗ · QUS 37✓ · UMS 55✗ · JAF 65✗ · SHJ 27✓	QUS, SHJ	2

Check: counts add to 26. Each route is counted from both ends, so routes = 26 ÷ 2 = **13** ✓

5. Step 2: run DFS from HUB

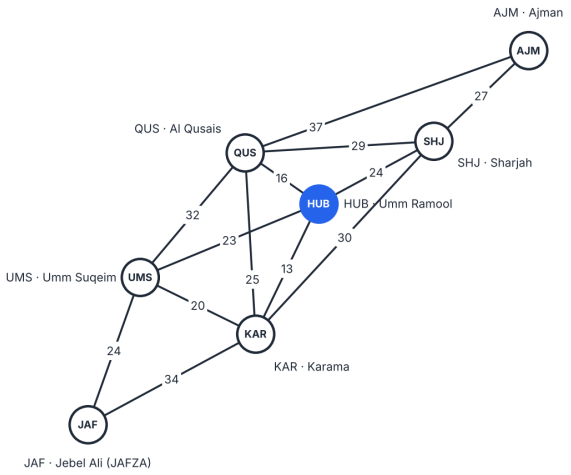
Rule: at the current branch, check its neighbours in matrix order (HUB, KAR, QUS, UMS, JAF, SHJ, AJM) and move to the **first unvisited** one (push it on the stack). If all its neighbours are already visited, **backtrack** (pop it) to the branch you came from. Stop when the stack is empty.

#	At	Neighbour checks	Action	Stack after	Visited count
1	-	start	**Visit HUB**	HUB	1
2	HUB	KAR: not visited → go	**Visit KAR**	HUB > KAR	2
3	KAR	HUB ✗ QUS: not visited → go	**Visit QUS**	HUB > KAR > QUS	3
4	QUS	HUB ✗ KAR ✗ UMS: not visited → go	**Visit UMS**	HUB > KAR > QUS > UMS	4
5	UMS	HUB ✗ KAR ✗ QUS ✗ JAF: not visited → go	**Visit JAF**	HUB > KAR > QUS > UMS > JAF	5
6	JAF	KAR ✗ UMS ✗ → nothing new	Backtrack (pop JAF)	HUB > KAR > QUS > UMS	5
7	UMS	HUB ✗ KAR ✗ QUS ✗ JAF ✗ → nothing new	Backtrack (pop UMS)	HUB > KAR > QUS	5
8	QUS	HUB ✗ KAR ✗ UMS ✗ SHJ: not visited → go	**Visit SHJ**	HUB > KAR > QUS > SHJ	6
9	SHJ	HUB ✗ KAR ✗ QUS ✗ AJM: not visited → go	**Visit AJM**	HUB > KAR > QUS > SHJ > AJM	7
10	AJM	QUS ✗ SHJ ✗ → nothing new	Backtrack (pop AJM)	HUB > KAR > QUS > SHJ	7
11	SHJ	HUB ✗ KAR ✗ QUS ✗ AJM ✗ → nothing new	Backtrack (pop SHJ)	HUB > KAR > QUS	7
12	QUS	HUB ✗ KAR ✗ UMS ✗ SHJ ✗ AJM ✗ → nothing new	Backtrack (pop QUS)	HUB > KAR	7
13	KAR	HUB ✗ QUS ✗ UMS ✗ JAF ✗ SHJ ✗ → nothing new	Backtrack (pop KAR)	HUB	7
14	HUB	KAR ✗ QUS ✗ UMS ✗ SHJ ✗ → nothing new	Backtrack (pop HUB)	(empty)	7

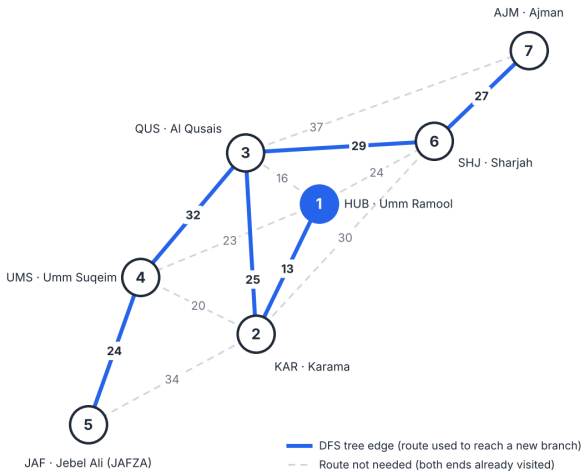
6. Result

Item	Result
Visited / total branches	**7 / 7 → CONNECTED**
Visit order	HUB → KAR → QUS → UMS → JAF → SHJ → AJM
DFS tree (routes used)	HUB-KAR, KAR-QUS, QUS-UMS, UMS-JAF, QUS-SHJ, SHJ-AJM (6 = 7 − 1)
Routes not needed by DFS	HUB-QUS, HUB-SHJ, HUB-UMS, KAR-JAF, KAR-SHJ, KAR-UMS, QUS-AJM
Work done	26 neighbour checks, 7 visits, 7 backtracks: O(V + E)

Route network at X = 38 min · 13 routes (drive time under 38 min)



Depth-first search from HUB · number = visit order



Times: Google Maps driving, depart Thu 5 Nov 2026 07:00, midpoint of typical range. Each pair used in both directions. Layout is schematic (follows real W-E / S-N order, not to scale).
Visit order: HUB > KAR > QUS > UMS > JAF > SHJ > AJM · All 7 visited: CONNECTED · No single point of failure: every branch has at least two routes

7. Findings

No single point of failure. Every branch has at least two routes, so if any one office closes, the other six stay connected. Ajman now has two routes, Sharjah (27 min) and Al Qusais (37 min). At X = 35 it had only Sharjah, so raising X to 38 removes that weak point.

Delivery from the hub is still slow to the far offices. Raising X to 38 does not restore the 38-minute direct routes. The fastest hub deliveries are Jebel Ali 47 min (via Karama or Umm Suqeim, a tie) and Ajman 51 min (via Sharjah). These are the candidates for upgrades in Tab 3.

Cross-check: networkx is_connected() = True at X = 38.